Ranjan Bassi

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EDUCATION

Dec 2020 – May 2025
Bhopal, India

Bachelors & Masters of Science, Physics
Indian Institute of Science Education and Research
CPI - 8.18 (on a 10 point scale)

May 2019
Chandigarh, India

12th Standard
New Public School
Grade 12, Science Stream

PROFESSIONAL EXPERIENCE

Aug 2024 – present
Bhopal

MS Thesis: Plasmonic interactions and dynamics in Transition Metal Dichalcogenides
Prof. Varadharajan Srinivasan, Prof. Sebastian Wüster
- Performed atomistic simulations of plasmonic dynamics in WSe_2 nanoflake heterostructures (bilayer, bowtie, and bowtie-bowtie configurations) using real-time Time-Dependent Density Functional Theory
- Investigated electric field enhancement, energy transfer and charge transfer phenomena within these nanostructures
- Developed a Python-based computational pipeline for analyzing charge transfer dynamics and a real-time electron density visualization and sampling tool

Jan 2024 – present
Bhopal, India

Developer, Litesoph
National Supercomputing Mission (under Dr. Varadharajan Srinivasan)
- Working on an RT-TDDFT toolkit to make light induced phenomenon accessible to researchers in a wide array of fields. (Project link)
- ***Code Agnostic Framework*:**
  Integrates multiple computational engines with an engine agnostic architecture (Octopus, GPAW, NWChem) supporting all widely used basis sets under one umbrella
- **Facilitating *studies in*:**
  Photochemistry and optoelectronic materials

May 2023 – Aug 2023
Hamilton, Canada

Mitacs Research Fellow
MacMaster University, Prof Timothy Field
- Applied Stochastic Differential Equations to Clarke's wireless communication model, correcting spectrum errors, and enhanced MIMO systems with Space-Time coding
- Conducted an extensive review of over eight academic papers to underpin model development and stochastic calculus applications. (Project report link)

May 2022 – Jul 2022
Bhopal, India

Research Intern, Spin Lab
IISER Bhopal, Dr Phani Kumar Peddibhotla
Determination Of Zero Field Electron Spin Resonance Spectrum For Nitrogen vacancy defects
- Executed simulations of magnetic nanoparticles using Python & Mathematica, mathematically modeling their magnetic fields across various conditions
- Studied & explored the techniques involved in using Nitrogen vacancy centers as nano sensors for magnetic fields
- Detected magnetic domain boundaries for various materials using the above techniques

Nov 2022 – Jan 2023
Bhopal, India

Team lead, International Physicists tournament
IISER Bhopal
- Worked on 3 open ended problems for IPT
- Lead a team of 7, wrote a report on the chalk problem (Project report link)

RESEARCH COMMUNICATION AND PRESENTATIONS

Integrated Toolkit for Simulations of Photo-induced Phenomena: An Excited State Dynamics Toolkit
Poster Presentation: 4th PSCES'24 (IISER Bhopal, Apr 8–10, 2024) & 8th InTeRaCTIONS Symposium (Chemistry Dept., IISER Bhopal, Mar 13–14, 2024)
Presented a Python toolkit facilitating accessible excited-state simulations (via customizable RT-TDDFT) for researchers studying photoinduced phenomena in computational photochemistry and optoelectronics

One-Day User Training Workshop on LITESOPH
Mastering the Layer Integrated Toolkit for Simulations of Photo-Induced Phenomena in Computational Chemistry
Co-delivered training (demos, hands-on, Q&A) on the LITESOPH photo-induced phenomena simulation toolkit at a CDAC (Center for development of advanced computing)-hosted workshop for materials/computational chemistry researchers (Guidance: Prof. Varadharajan)

Probing Photo-Excited Phenomena in TMDC Nanoflakes via TDDFT
Best Poster Award, 2025
Poster Presentation: 9th InTeRaCTIONS Symposium (Chemistry Dept., IISER Bhopal); **Won Best Poster Award** for thesis work (TMDC nanoplasmonics)

PUBLICATIONS

Plasmonic Interactions in Transition-Metal Dichalcogenide Nanoflakes
Master's Thesis, Indian Institute of Science Education and Research, Bhopal (Expected Submission: 13th April 2025)

Manuscript based on MS Thesis Research
Manuscript in Preparation

SELF-GUIDED COMPUTATIONAL PROJECTS

Ray Tracing in Continuous media

Gaussian beam through Fabry-Perot cavity

PROGRAMMING LANGUAGES

Mathematica | Python | LaTeX | C

TOOLS AND SCIENTIFIC CODES

Linux | Git | HPC | GPAW | Octopus

CO-CURRICULAR INTERESTS

Personal blog where I talk about science and philosophy

AWARDS

Mitacs GRI Scholar

Selected as the lead developer on a national supercomputing mission project, LITESOPH

Best Poster Award 2025, INTERACTIONS, In-house symposium, Department of Chemistry, IISER Bhopal